Listing 5.1 `first_function.py` (API call)

```
response = client.chat.completions.create(
        model="gpt-4-1106-preview",
        messages=[{"role": "system",
                   "content": "You are a helpful assistant."},
                  {"role": "user", "content": user_message}],
        temperature=0.7,
        tools=[     #1
            {
                "type": "function",     #2
                "function": {
                    "name": "recommend",
                    "description": "Provide a … topic.",     #3
                    "parameters": {
                        "type": "object",     #4
                        "properties": {
                            "topic": {
                                "type": "string",
                                "description":
                                   "The topic,… for.",     #5
                            },
                            "rating": {
                                "type": "string",
                                "description":
                              "The rating … given.",    #5
                                "enum": ["good",
                                         "bad",
                                         "terrible"]     #6
                                },
                        },
                        "required": ["topic"],
                    },
                },
                }
            ]
        )
```

#1 New parameter called tools
#2 Sets the type of tool to function
#3 Provides an excellent description of what the function does
#4 Defines the type of parameters for input; an object represents a JSON document.
#5 Excellent descriptions for each input parameter
#6 You can even describe in terms of enumerations.

To see how this works, open Visual Studio Code (VS Code) to the book's source code folder: `chapter_4/first_function.py`. It's a good practice to open the relevant chapter folder in VS Code to create a new Python environment and install the `requirements.txt` file. If you need assistance with this, consult appendix B.

Before starting, correctly set up an `.env` file in the `chapter_4` folder with your API credentials. Function calling is an extra capability provided by